Antioxidant, antimutagenic, and antitumor effects of pine needles (Pinus densiflora).

Pine needles (Pinus densiflora Siebold et Zuccarini) have long been used as a traditional health-promoting medicinal food in Korea. To investigate their potential anticancer effects, antioxidant, antimutagenic, and antitumor activities were assessed in vitro and/or in vivo. Pine needle ethanol extract (PNE) significantly inhibited $Fe(2+)$-induced lipid peroxidation and scavenged 1,1-diphenyl- 2-picrylhydrazyl radical in vitro. PNE markedly inhibited mutagenicity of 2-anthramine, 2-nitrofluorene, or sodium azide in Salmonella typhimurium TA98 or TA100 in Ames tests. PNE exposure effectively inhibited the growth of cancer cells (MCF-7, SNU-638, and HL-60) compared with normal cell (HDF) in 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide assay. In in vivo antitumor studies, freeze-dried pine needle powder supplemented (5%, wt/wt) diet was fed to mice inoculated with Sarcoma-180 cells or rats treated with mammary carcinogen, 7,12-dimethylbenz[a]anthracene (DMBA, 50 mg/kg body weight). Tumorigenesis was suppressed by pine needle supplementation in the two model systems. Moreover, blood urea nitrogen and aspartate aminotransferase levels were significantly lower in pine needle-supplemented rats in the DMBA-induced mammary tumor model. These results demonstrate that pine needles exhibit strong antioxidant, antimutagenic, and antiproliferative effects on cancer cells and also antitumor effects in vivo and point to their potential usefulness in cancer prevention.